

Project Proposal: Secure Network Infrastructure for GreenTech Solutions

Client: GreenTech Solutions **Project Title:** Design and Implementation of a Secure, Scalable, and Cost-Effective Network Infrastructure

1. Executive Summary

This proposal outlines the design and implementation plan for a secure network infrastructure for GreenTech Solutions, addressing the need for secure inter-branch communication, reliable internet access, and robust security policies. The solution leverages Fortinet's industry-leading security fabric, specifically the **FortiGate 60F Next-Generation Firewalls (NGFWs)** and **FortiManager** for centralized management, ensuring a scalable and cost-effective deployment.

2. Network Design and Topology

The proposed network architecture is a two-site hub-and-spoke model, with the Headquarters (HQ) acting as the central hub and the Remote Office (RO) as the spoke. A secure **IPsec Site-to-Site VPN tunnel** will connect the two locations.

2.1. Network Topology Diagram (Conceptual)

The network will consist of the following key components at each site:

Component	Location	Role
FortiGate 60F	HQ & RO	Edge Firewall, Router, VPN Endpoint, and Security Gateway.
Managed Switch	HQ & RO	Layer 2 device for VLAN segmentation and connectivity.
Internal Hosts	HQ & RO	End-user devices (PCs, Servers, Printers).

HQ Site: FortiGate connects to the ISP (WAN) and the internal Managed Switch. The switch distributes connectivity to internal hosts across segmented VLANs. **RO Site:** FortiGate

connects to the ISP (WAN) and the internal Managed Switch, which connects to the RO hosts.

2.2. IP Addressing Scheme

A private, non-overlapping IP addressing scheme is defined to ensure proper routing and VPN functionality.

Location	Network	Subnet Mask	Gateway (FortiGate Internal Interface)	Purpose
HQ - Management	192.168.10.0	255.255.255.240 (/28)	192.168.10.1	Network devices, servers, and IT staff.
HQ - Users/Data	192.168.20.0	255.255.255.0 (/24)	192.168.20.1	Employee workstations and primary data access.
HQ - Guest/IoT	192.168.30.0	255.255.255.0 (/24)	192.168.30.1	Isolated network for guest access and IoT devices.
RO - Users/Data	192.168.40.0	255.255.255.0 (/24)	192.168.40.1	Remote office employee workstations and data access.
WAN (Simulated)	HQ: 203.0.113.1	N/A	N/A	Public IP for VPN termination.
WAN (Simulated)	RO: 198.51.100.1	N/A	N/A	Public IP for VPN termination.

3. Required Equipment List

The following equipment is required for the implementation:

Item	Model	Quantity	Estimated Unit Cost (USD)	Total Cost (USD)	Notes
Firewall	FortiGate 60F	2	\$700	\$1,400	Next-Generation Firewall for each branch.
Security Subscription	FortiGuard UTP (1-Year)	2	\$400	\$800	Unified Threat Protection (AV, IPS, Web Filtering, etc.).
Switch (Managed)	Cisco Catalyst 2960 (Simulated)	2	\$300	\$600	Layer 2 switch with VLAN support.
Central Management	FortiManager VM (Trial/Demo)	1	\$0	\$0	Virtual Appliance for centralized policy management.
Cabling & Accessories	CAT6, Racks, UPS (Estimated)	1 Lot	\$500	\$500	Miscellaneous hardware and power protection.
Total Hardware & Software Cost				\$3,300	

4. Security Policies and Access Control Strategy

The security strategy is based on the principle of **Least Privilege** and **Defense-in-Depth**, enforced by the FortiGate NGFWs.

4.1. VLAN Segmentation

VLANs are used to segment the network, limiting the blast radius of a security incident and controlling internal traffic flow.

VLAN ID	Network	Policy
10 (Management)	192.168.10.0/28	Highly restricted access; only IT staff and necessary services.
20/40 (Users/Data)	192.168.20.0/24, 192.168.40.0/24	Standard employee access; full access to internal resources and filtered internet.
30 (Guest/IoT)	192.168.30.0/24	Internet-only access; completely isolated from internal networks (VLAN 10, 20, 40).

4.2. FortiGate Firewall Policies (Key Examples)

Policy Name	Source Interface/Zone	Destination Interface/Zone	Source Address	Destination Address	Service
1. Inter-Branch VPN	Internal (VLAN 20, 40)	VPN Tunnel	192.168.20.0/24, 192.168.40.0/24	192.168.40.0/24, 192.168.20.0/24	RDP, SIP, Ping
2. Outbound Internet	Internal (VLAN 20, 40)	WAN	Internal Subnets	All	HTTP, HTTPS, DNS
3. Guest Internet	Internal (VLAN 30)	WAN	192.168.30.0/24	All	HTTP, HTTPS, DNS
4. Inter-VLAN Deny	Internal (VLAN 30)	Internal (VLAN 10, 20)	All	All	All
5. Default Deny	Any	Any	Any	Any	All

5. FortiGate and FortiManager Usage Plan

5.1. FortiGate Usage

The FortiGate 60F will be configured to provide the following security services:

- **Site-to-Site VPN:** Establish a secure IPsec tunnel between HQ and RO.
- **Unified Threat Protection (UTP):** Utilize FortiGuard services (Intrusion Prevention System, Antivirus, Web Filtering, Application Control) on all outbound traffic.
- **VLAN Routing:** Act as the default gateway for all internal VLANs and enforce inter-VLAN policies.
- **Logging and Reporting:** Forward all logs to FortiManager for centralized analysis and compliance.

5.2. FortiManager Usage

FortiManager will be deployed as a central management platform to achieve the following success criteria:

- **Centralized Policy Management:** Create and manage a single, consistent set of firewall policies that are deployed to both the HQ and RO FortiGate devices. This ensures consistency and reduces configuration errors.
- **Device Provisioning:** Simplify the initial setup and ongoing configuration changes for both FortiGate units.
- **Firmware Management:** Centrally manage and schedule firmware upgrades to maintain security and stability across both devices.
- **Monitoring and Reporting:** Provide a single pane of glass for network visibility, security event monitoring, and compliance reporting.

6. Implementation Timeline

The project is estimated to take 4 weeks from the initial design approval to final documentation.

Phase	Duration	Key Activities	Deliverable
Phase 1: Design & Planning	1 Week	Finalize topology, IP scheme, equipment list, and security policies.	Project Proposal (This Document)

Phase 2: Procurement & Staging	1 Week	Order and receive equipment. Initial configuration of FortiManager VM.	Equipment Receipt, FortiManager Ready.
Phase 3: Implementation & Configuration	1 Week	Install FortiGates and Switches. Configure VLANs, VPN tunnel, and firewall policies (via FortiManager).	Simulated Network Configuration
Phase 4: Testing & Documentation	1 Week	Test VPN connectivity, internet access, and security policies. Finalize all configuration and test documentation.	Final Documentation & Test Results

7. Financial Proposal

The financial proposal covers the one-time costs for hardware, software subscriptions, and professional services for the design and implementation.

Description	Cost (USD)	Notes
Hardware & Software (Section 3 Total)	\$3,300	FortiGate 60F (x2), UTP Subscriptions, Switches, Accessories.
Professional Services (Design & Implementation)	\$4,500	45 hours @ \$100/hour for network design, configuration, testing, and documentation.
Contingency (10%)	\$780	Buffer for unforeseen costs or additional requirements.
Total Project Cost	\$8,580	Total cost for a secure, fully implemented network.

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